

Quiz (Carolina Reaper)

- Q1.** A shipping company has 2^3 times more trucks than cars. If they have 8 cars, how many trucks do they have?
- (A) 16
(B) 24
(C) 32
(D) 40
(E) 48
- Q2.** Simplify the expression: $(2^2)^3$
- (A) 2^5
(B) 2^6
(C) 2^8
(D) 2^{12}
(E) 2^{20}
- Q3.** A traffic flow model uses the equation $v = 2^t$ to estimate the velocity of cars, where t is time in hours. What is the velocity after 4 hours?
- (A) 4
(B) 8
(C) 16
(D) 32
(E) 64
- Q4.** A logistics company has 3^2 times more warehouses than distribution centers. If they have 9 distribution centers, how many warehouses do they have?
- (A) 18
(B) 27
(C) 36
(D) 45
(E) 81
- Q5.** Simplify the expression: $\frac{2^4}{2^2}$
- (A) 2^2
(B) 2^4
(C) 2^6
(D) 2^8
(E) 2^{16}
- Q6.** A delivery company uses the equation $d = 2^{n-1}$ to estimate the number of delivery routes, where n is the number of drivers. What is the number of delivery routes if there are 5 drivers?
- (A) 8
(B) 16
(C) 32
(D) 64
(E) 128
- Q7.** Simplify the expression: $(3^2)^2$
- (A) 3^4
(B) 3^6
(C) 3^8
(D) 3^{12}
(E) 3^{16}
- Q8.** A transportation company has 4^2 times more buses than trains. If they have 16 buses, how many trains do they have?
- (A) 1
(B) 2
(C) 4
(D) 8
(E) 16

Open Ended Questions (FRQ)

- Q9.** A shipping company uses the equation $t = 2^{d-1}$ to estimate the time it takes to deliver a package, where d is the distance in miles. If it takes 8 hours to deliver a package 16 miles away, what is the value of d if it takes 32 hours to deliver a package?
- Q10.** Simplify the expression: $\frac{(2^3)^2}{(2^2)^3}$ and explain your steps.

Chef's Tips

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- Tip 1: Encourage students to use the exponent rules to simplify expressions.
- Tip 2: Remind students to use the correct units when solving problems.
- Tip 3: Emphasize the importance of checking calculations for accuracy.